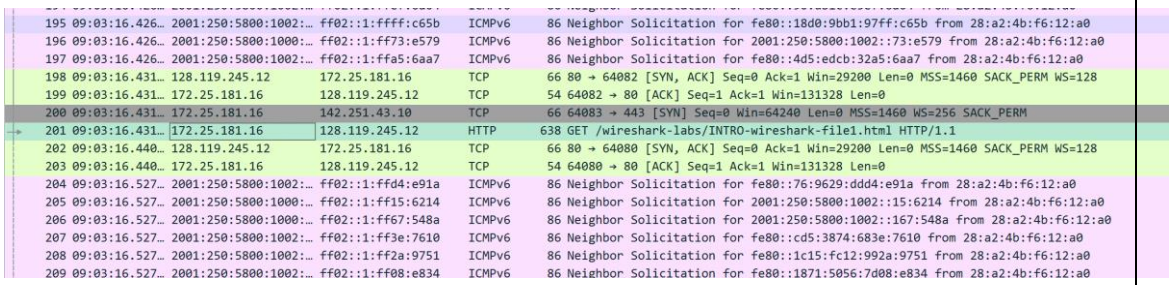
山东大学 计算机科学与技术 学院

计算机网络 课程实验报告

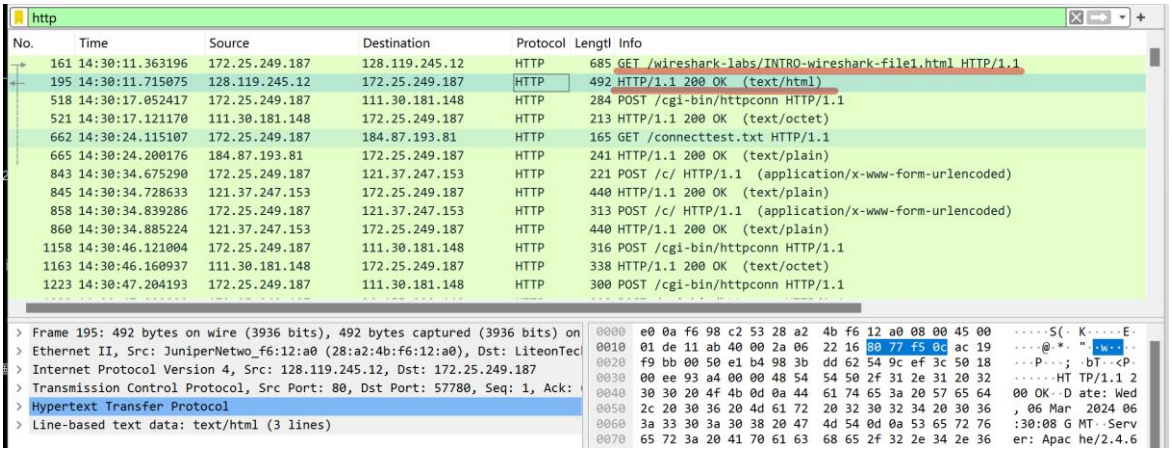
学号：202200130077	姓名：薛润泽	班级：智能班
实验题目：计算机网络实验一		
实验学时：2	实验日期：2024.3.6	
实验目的：熟悉 Wireshark，启动运行 Wireshark 并探索他的功能		

实验结果：

1. Protocal:TCP,ICMPv6,HTTP

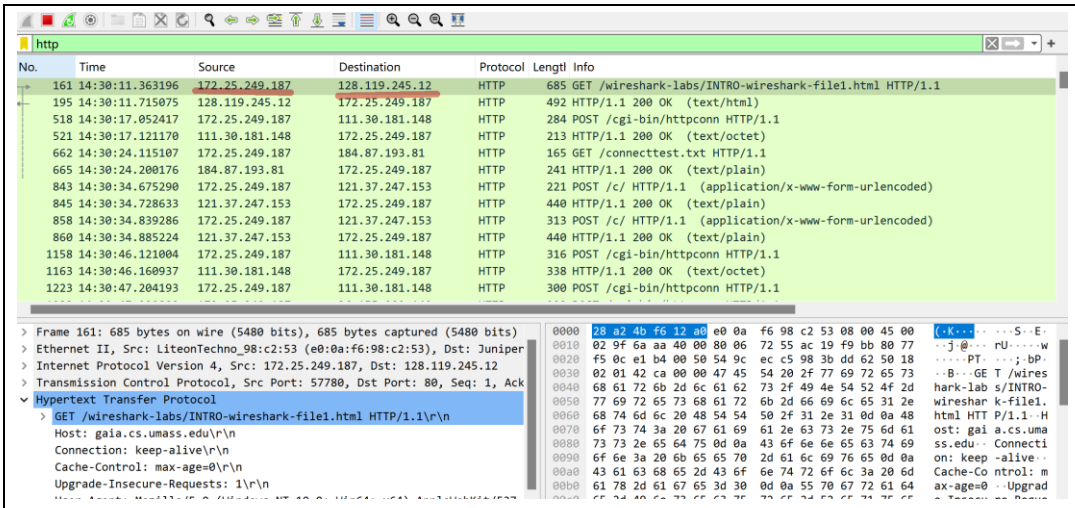


2. 0.351879 s



3. gaia.cs.umass.edu 互联网地址：128.119.245.12

我的互联网地址：172.25.249.187



4. 打印文件截屏如下

```
No.     Time               Source                Destination           Protocol Length Info
    161 14:30:11.363196    172.25.249.187        128.119.245.12        HTTP     685    GET /wireshark-labs/INTRO-wireshark-file1.html
HTTP/1.1
Frame 161: 685 bytes on wire (5480 bits), 685 bytes captured (5480 bits) on interface \Device\NPF_{D704B264-FD7E-4F03-BFA1-33646389BD21},
id 0
Ethernet II, Src: LiteonTechno_98:c2:53 (e0:0a:f6:98:c2:53), Dst: JuniperNetwo_f6:12:a0 (28:a2:4b:f6:12:a0)
Internet Protocol Version 4, Src: 172.25.249.187, Dst: 128.119.245.12
Transmission Control Protocol, Src Port: 57780, Dst Port: 80, Seq: 1, Ack: 1, Len: 631
Hypertext Transfer Protocol
    GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1\r\n
    Host: gaia.cs.umass.edu\r\n
    Connection: keep-alive\r\n
    Cache-Control: max-age=0\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/122.0.0.0 Safari/537.36 Edg/
122.0.0.0\r\n
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-
exchange;v=b3;q=0.7\r\n
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: zh-CN,zh;q=0.9,en;q=0.8,en-GB;q=0.7,en-US;q=0.6\r\n
    If-None-Match: "51-612f813182301"\r\n
    If-Modified-Since: Wed, 06 Mar 2024 06:29:01 GMT\r\n
    \r\n
    [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html]
    [HTTP request 1/1]
    [Response in frame: 195]
No.     Time               Source                Destination           Protocol Length Info
    195 14:30:11.715075    128.119.245.12        172.25.249.187        HTTP     492    HTTP/1.1 200 OK  (text/html)
Frame 195: 492 bytes on wire (3936 bits), 492 bytes captured (3936 bits) on interface \Device\NPF_{D704B264-FD7E-4F03-BFA1-33646389BD21},
id 0
Ethernet II, Src: JuniperNetwo_f6:12:a0 (28:a2:4b:f6:12:a0), Dst: LiteonTechno_98:c2:53 (e0:0a:f6:98:c2:53)
Internet Protocol Version 4, Src: 128.119.245.12, Dst: 172.25.249.187
Transmission Control Protocol, Src Port: 80, Dst Port: 57780, Seq: 1, Ack: 632, Len: 438
Hypertext Transfer Protocol
    HTTP/1.1 200 OK\r\n
    Date: Wed, 06 Mar 2024 06:30:08 GMT\r\n
    Server: Apache/2.4.6 (CentOS) OpenSSL/1.0.2k-fips PHP/7.4.33 mod_perl/2.0.11 Perl/v5.16.3\r\n
    Last-Modified: Wed, 06 Mar 2024 06:30:02 GMT\r\n
    ETag: "51-612f816b55935"\r\n
    Accept-Ranges: bytes\r\n
    Content-Length: 81\r\n
    Keep-Alive: timeout=5, max=100\r\n
    Connection: Keep-Alive\r\n
    Content-Type: text/html; charset=UTF-8\r\n
    \r\n
    [HTTP response 1/1]
    [Time since request: 0.351879000 seconds]
    [Request in frame: 161]
    [Request URI: http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html]
    File Data: 81 bytes
Line-based text data: text/html (3 lines)
```

问题及收获：

在使用 Wireshark 进行抓包的过程中，访问有关网址反现其响应信号为 Not Modified 而不是预期的 OK。经过多次尝试，发现更换网络环境可以实现预期。